

NORMALIZATION

①

StudentInfo

stdID	Name	Age	Brcode	BrName	HodName
1	A	19	101	CS	XYZ
2	B	18	101	CS	XYZ
3	C	19	101	CS	XYZ
4	D	20	102	ECE	ABC
5	E	21	102	ECE	ABC
6	F	19	103	Mech	DEF

Redundancy:- Duplicate values.

Anomalies:-

(1) Insertion anomaly - when certain data or attribute cannot be inserted into database without the presence of other data.

eg- suppose branch civil is to be added. And we can't add that without having some student.

(2) Deletion anomaly:- If we want to delete some unwanted data and it cause to delete some wanted data.

BY:- Dr. DEEPAK
AHLAWAT

(3) Updation Anamoly :- modify some data in multiple places.

Solution:- To create or decompose the tables.

Don't put all information in one table, instead use separate tables.

StudInfo

StudID	Name	Age	BeCode
1	A	19	101
2	B	18	101
3	C	19	101
4	D	20	102
5	E	21	102
6	F	19	103

BeTable

BeCode	BeName	HoDName
101	CS	XYZ
102	ECE	ABC
103	Mech	DEF

—————x—————x—————x—————x—————

(3)

Normalization is used to reduce redundancy (same data-duplicate copies) and solve anomalies.

1NF (1st Normal form)

In a table every attribute is atomic (means no multivalued attribute and no composite attribute).

We decompose the table into several tables, so as to establish the normalisation.

eg:- Enrollment table

studID	course
1	C, C++, Java

→ This is a multivalued attribute

Composite attribute: - Like Address - 140, COLET, Chitkara University

It has a structure. Its a structure combining three meaningful attributes in this case.

When a table is in 1NF, then languages like SQL fetches the table completely.

eg:- Convert table in 1NF

Sid	SName	course
1	A	C, C++
2	B	C++, Java
3	C	C, C++, Java

(4)

Sid	SName	Course
1	A	C
1	A	C++
2	B	C++
2	B	Java
3	C	C
3	C	C++
3	C	Java

Similarly, in case of composite attribute

Sid	SName	Address
1	A	488, Delhi, 110001
2	B	528, Mumbai, 400001
3	C	011, Chandigarh, 140901

Alter decompose,

Sid	SName	HNo	City	Pincode
1	A	488	Delhi	110001
2	B	528	Mumbai	400001
3	C	011	Chandigarh	140901

Now, we can create queries.

(*) Thus, when we have multivalued attribute then go for a new row for each attribute. But if we have composite attribute, then go for a new column.

2NF (2nd Normal Form)

A table is in 2NF, if there is no partial dependency.

$X \rightarrow a$ is a partial dependency, if X is a proper subset of some candidate key and a is a non-prime attribute.

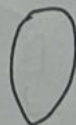
Prime attribute
or

Key attribute



Non-prime
or

Non-key attribute



eg:-

(1) R(ABCD)

$A \rightarrow B$

$C \rightarrow D$

(1) AC is the candidate key

$\begin{pmatrix} A \\ C \end{pmatrix}$ - Prime attribute

$\begin{pmatrix} B \\ D \end{pmatrix}$ - Non-prime attribute

Now, check for partial dependency

$A \rightarrow B$ — Non-prime
/
proper
subset of candidate key

So, this is a partial dependency
and thus the table is not in 2NF.

⊗ Atleast one partial dependency is sufficient
to tell that the table is not in 2NF.

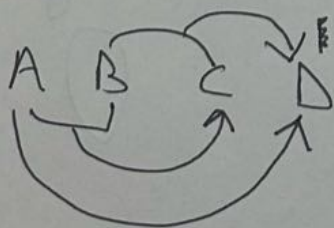
eg:- $R(ABCD)$

$AB \rightarrow C$

$BC \rightarrow D$

$A \rightarrow D$

Solution:-



A and B are essential

$(AB)^+ = ABCD$ —

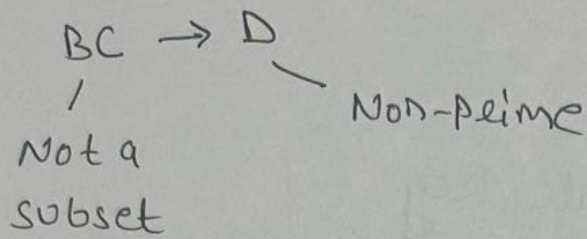
AB is the candidate key

check for partial dependency

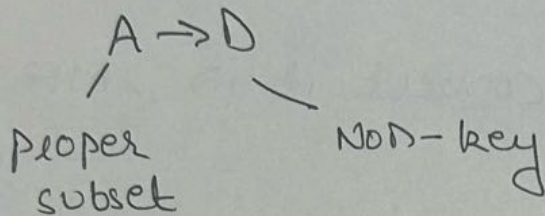
Not a proper
subset — $AB \rightarrow C$ — Non-prime

Not a partial Dependency

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so, Not a partial dependency



This is a partial dependency

Thus, not in 2NF.

Example:-

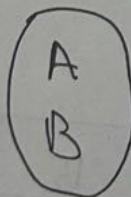
$R(ABCD)$

$AB \rightarrow D$

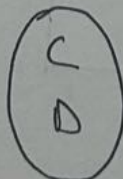
$B \rightarrow C$

Solution

AB is candidate key



Prime



Non-prime

$B \rightarrow C$

Partial Dependency

Not in 2NF

Eg:- $R(ABC)$

$B \rightarrow C$

Sol:-

AB is candidate key

Partial dependency

Not in 2NF

Suppose, we want to convert it in 2NF

Decompose

$R_1(AB)$ $R_2(BC)$

Eg:-

R		
A	B	C
a	1	x
b	2	y
a	3	z
c	3	z
d	3	z
e	3	z

R_1

A	B
a	1
b	2
a	3
c	3
d	3
e	3

R_2

B	C
1	x
2	y
3	z

Eg:- Find whether it is in 2NF or not. Then decompose the tables in 2NF. (9)

Ans

eg:- $R(ABCDE)$

$AB \rightarrow C$

$D \rightarrow E$

Sol:-

A B C D E
└──┬──┘ └──┘
 ↑ ↑

ABD is essential

$(ABD)^+ = ABDCDE = \text{Candidate key}$

Now:-

$AB \rightarrow C$
/ \
proper subset Non-key

It is partial dependency

$D \rightarrow E$
/ \
proper subset Non-key

It is PD

The R is not in 2NF

Now, Decompose

├ $R_1(ABC)$
├ $R_2(DE)$
└ $R_3(ABD)$

Ex

R(A B C D E)

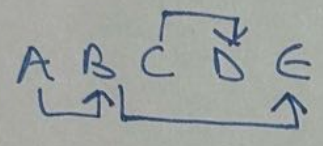
A → B

B → E

C → D

Sol

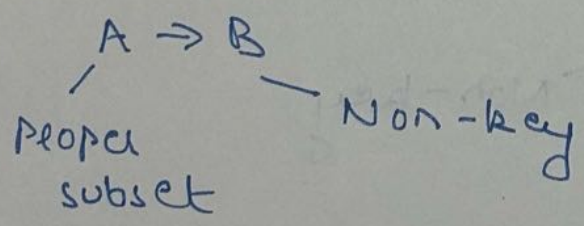
Find CK



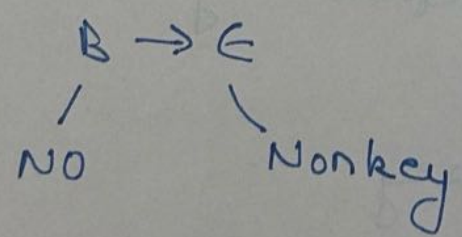
AC is essential

$$(AC)^+ = ACBED = CK$$

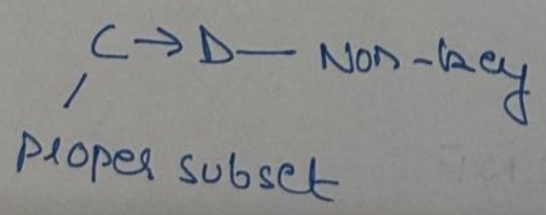
Now,



It is PD



It is OK



It is PD

Not in 2NF, Decompose

- R1(A B E)
- R2(C D)
- R3(A C)

Eg:- $R(ABCDEFGHIJ)$

$AB \rightarrow C$

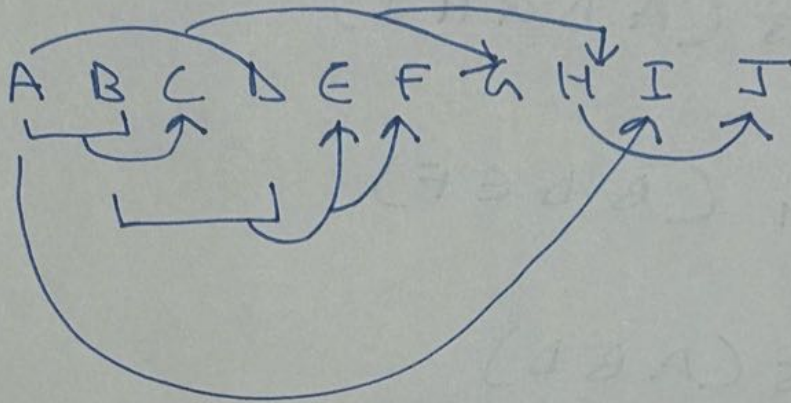
$AD \rightarrow GH$

$BD \rightarrow EF$

$A \rightarrow I$

$H \rightarrow J$

Sol:-



ABD is essential

$$(ABD)^+ = ABCDGH E F I J = CK$$

check for PD

$AB \rightarrow C$ (PD)

$AD \rightarrow GH$ (PD)

$BD \rightarrow EF$ (PD)

$A \rightarrow I$ (PD)

$H \rightarrow J$ (OK)

Not in 2NF

Decompose: -

- $R_1 (ABC)$
- $R_2 (AI)$
- $R_3 (ADGHI)$
- $R_4 (BDEF)$
- $R_5 (ABD)$

— $\times$ — $\times$ — $\times$ — $\times$

Trivial / Non-Trivial FD:-

(13)

An FD $\alpha \rightarrow \beta$, where $\beta \subseteq \alpha$ called a trivial FD, it always holds good.

An FD $x \rightarrow y$, where $y \not\subseteq x$ is non-trivial FD.

An FD $x \rightarrow y$, where $x \cap y = \emptyset$, is completely non-trivial FD.

Transitive FD:- A FD from $\alpha \rightarrow \beta$ is called transitive if $\alpha, \beta \in \text{Non-Prime}$.

3NF:- A Relation R is said to be in 3NF, if it has no partial dependency and no transitive dependency.

Other form of definition:-

A table is in 3NF

if

(1) It is in 1NF and 2NF

(2) For every non-trivial FD

$$x \rightarrow a$$

x is either a superkey

or, a is a prime attribute.

eg:- $R(ABC)$

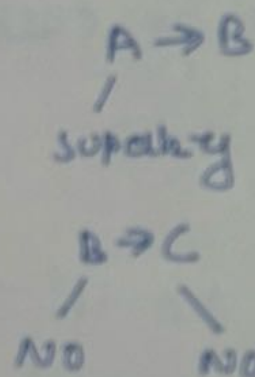
$A \rightarrow B$

$B \rightarrow C$

Ans Key is A

A
Prime
attribute

B, C
Non-prime



Both condition must be true, so it is not in 3NF.

eg:- $R(ABCD)$

$AB \rightarrow C$

$B \rightarrow C$

$C \rightarrow B$

Sol:-



A and D are essential

So, keys are BAD and CAD

A, B, C, D
Prime

$\emptyset$
Non Prime

So, on right side, there is always a prime.

The relation is in 3NF.

⊗ Some non-prime attributes when transitively dependent on candidate key then that table can't be in 3NF.

⊗ Superkey doesn't determine the candidate key, but candidate key can determine a superkey.

eg: R(ABC)
 $A \rightarrow B$
 $B \rightarrow C$

Sol

A	B	C
a	1	x
b	1	x
c	1	x
d	2	y
e	2	y
f	2	y
g	3	z
h	3	z

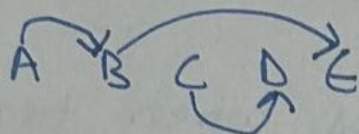
Decompose

A	B
a	1
b	1
c	1
d	2
e	2
f	2
g	3
h	3

B	C
1	x
2	y
3	z

Ques

R(ABCDE)

 $A \rightarrow B$ $B \rightarrow E$ $C \rightarrow D$ Ans

AC essential

 $(AC)^+ = ABECD = \text{Cand. Key}$

Now, check FD's

$A \rightarrow B$
 / \
 Proper subset Non prime
 (Not a superkey)

So, Not in 3NF

To convert it in 3NF, Decompose

$R_1(A, B, E)$ — $R_{11}(A, B)$
 $R_{12}(B, E)$
 $R_2(C, D)$
 $R_3(A, C)$

Ans R(ABCDEFGHIJ)

(17)

$AB \rightarrow C$

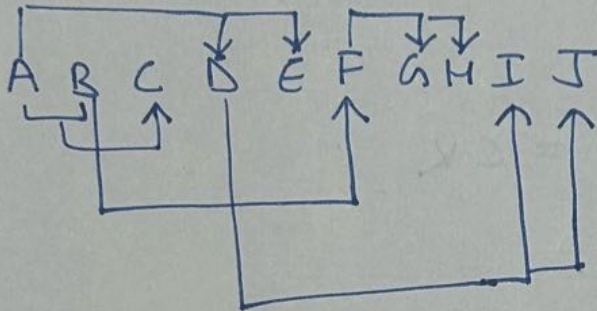
$A \rightarrow DE$

$B \rightarrow F$

$F \rightarrow GH$

$D \rightarrow IJ$

Ans



AB is essential

$(AB)^+ = ABCDEFGHIJ = CK$

$AB \rightarrow C$ ✓
Super key

X $A \rightarrow DE$

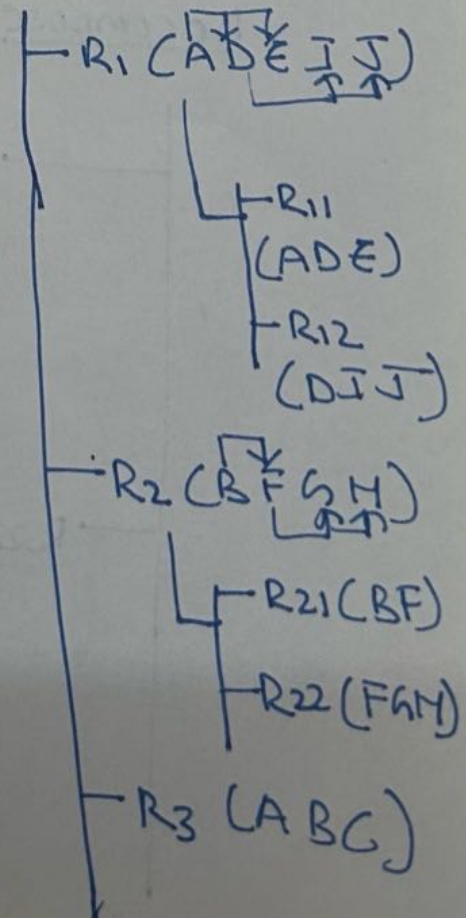
X $B \rightarrow F$

X $F \rightarrow GH$

X $D \rightarrow IJ$

Not in 3NF

Now, Decompose



ans

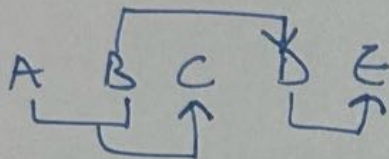
R (A B C D E)

$AB \rightarrow C$

$B \rightarrow D$

$D \rightarrow E$

Ans



A, B essential

$(AB)^+ = ABCDE = CK$

~~Dec~~

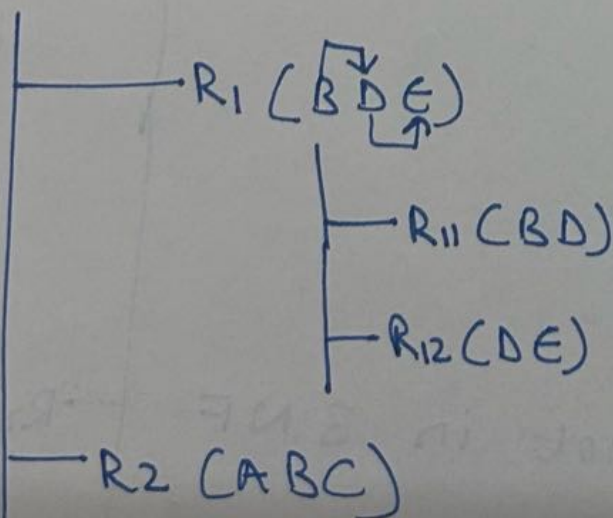
$AB \rightarrow C \checkmark$

$B \rightarrow D \times$

$D \rightarrow E \times$

Not in 3NF

Decompose



Ques

R(ABCDEFGHIJ)

$AB \rightarrow C$

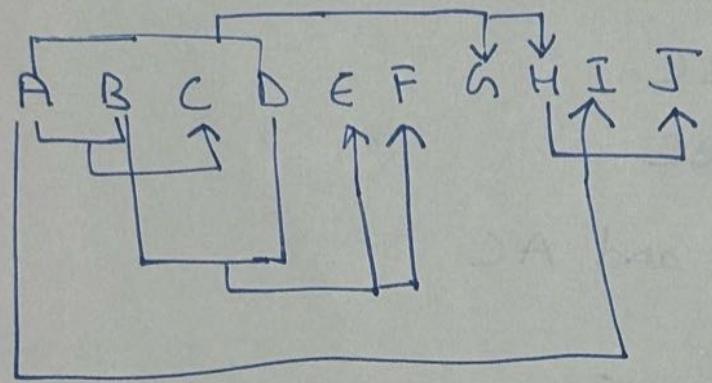
$AD \rightarrow GH$

$BD \rightarrow EF$

$A \rightarrow I$

$H \rightarrow J$

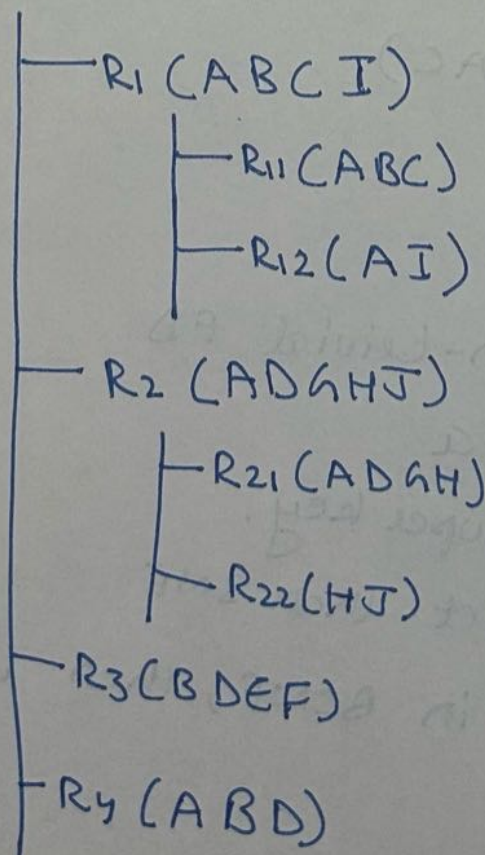
Ans



ABD is CK

All FD fails for 3NF, so not in 3NF

Decompose



BCNF (Boyce - Codd Normal Form) :-

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$$R(\overline{A} \overline{B} \overline{C})$$

$$AB \rightarrow C$$

$$C \rightarrow B$$

A is essential

$$A^+ = A$$

$$(AB)^+ = ABC$$

$$(AC)^+ = ABC$$

Two keys AB and AC

$$AB \rightarrow C \quad \checkmark$$

$$C \rightarrow B \quad \times$$

$$\begin{array}{|l} \hline R_1(CB) \\ \hline R_2(AC) \\ \hline \end{array}$$

So, IN BCNF

For every non-trivial FD

$$X \rightarrow a$$

X is a super key.

It is more strict than 3NF

If your table is in BCNF, then it is automatically in 3NF.

Exercise

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Ques 1 R(A B C D E F G H)

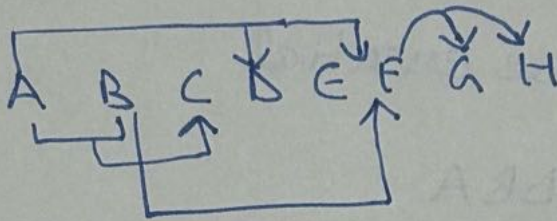
$AB \rightarrow C$

$A \rightarrow DE$

$B \rightarrow F$

$F \rightarrow GH$

Ans



A and B are essential

$(AB)^+ = ABCDEFGH$

$AB = C$

Now, check for BCNF
 $AB \rightarrow C$ ✓

$A \rightarrow DE$ X

check for 3NF

$A \rightarrow DE$ X

check for 2NF

$A \rightarrow DE$ X

The relation is in 1NF. It is not in 2NF, 3NF, or BCNF.

ques 2

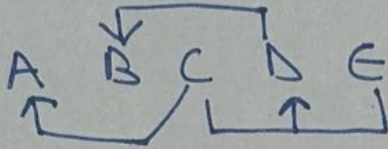
R(A B C D E)

$CE \rightarrow D$

$D \rightarrow B$

$C \rightarrow A$

Ans



C and E are essential

$(CE)^+ = CEDBA$

$CE = CK$

check for BCNF

$CE \rightarrow D$ ✓

$D \rightarrow B$ ✗

check for 3NF

$D \rightarrow B$ ✗

check for 2NF

$D \rightarrow B$ ✓

$C \rightarrow A$ ✗

It is in 1NF.

Ques 3.

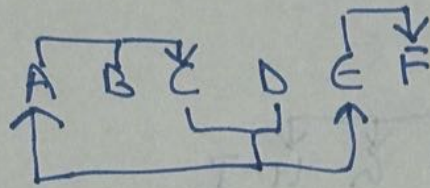
R(ABCDEF)

$AB \rightarrow C$

$DC \rightarrow AE$

$E \rightarrow F$

Ans



B and D are essential

$$(BD)^+ = BD$$

check A

$$(ABD)^+ = ABCDEF$$

check C

$$(BCD)^+ = BCDAEF$$

Two candidate keys ABD and BCD

Now, check for BCNF

$AB \rightarrow C$ X

check for 3NF

$AB \rightarrow C$ ✓

$DC \rightarrow AE$ X

check for 2NF

$DC \rightarrow AE$ X

It is in 1NF.

Ans 4

R(A B C D E F G H I)

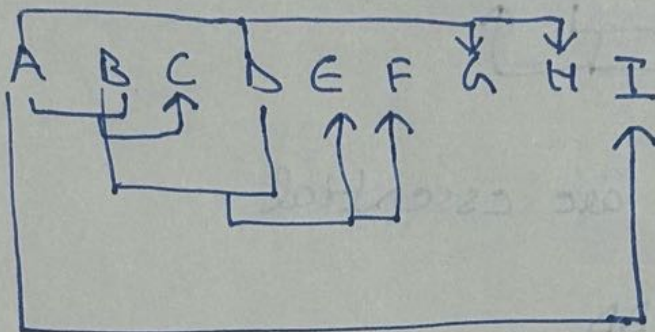
$AB \rightarrow C$

$BD \rightarrow EF$

$AD \rightarrow GH$

$A \rightarrow I$

Ans 4



A, B, D are essential

$(ABD)^+ = ABCDEFGHI$ ✓

$ABD = CK$

check for BCNF

$AB \rightarrow C$ X

check for 3NF

$AB \rightarrow C$ X

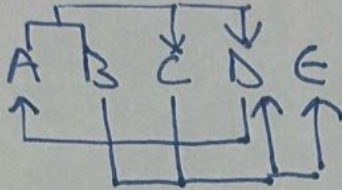
check for 2NF

$AB \rightarrow C$ X

It is in 1NF.

Ans 5

R(ABCDE)

 $AB \rightarrow CD$ $D \rightarrow A$ $BC \rightarrow DE$ Ans 5

B is essential

 $B^+ = B$ $(AB)^+ = ABCDE \checkmark$ $(BC)^+ = BCDEA \checkmark$ $(BD)^+ = BDACE \checkmark$

Three keys - AB, BC, BD

check for BCNF

 $AB \rightarrow CD \checkmark$ $D \rightarrow A \times$

check for 3NF

 $D \rightarrow A \checkmark$ $BC \rightarrow DE \checkmark$

The relation is in 3NF.

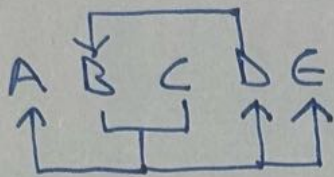
ans 6

RCAB CDE)

$BC \rightarrow ADE$

$D \rightarrow B$

Ans 6



C is essential

$C^+ = C$

$(AC)^+ = AC$

$(BC)^+ = BC ADE$ ✓

$(DC)^+ = DCBAE$ ✓

So,

Keys = BC and DC

Check for BCNF

$BC \rightarrow ADE$ ✓

$D \rightarrow B$ ✗

Check for 3NF

$D \rightarrow B$ ✓

It is in 3NF.

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BY:-
DR. DEEPAK AHLAWAT